

• Data Loading

Q. one-hot encoding

Let $Y = \begin{bmatrix} 2 \\ 0 \\ 3 \\ 1 \end{bmatrix} \xrightarrow{\text{flatten}} Y_{\text{flat}} = [2 \ 0 \ 3 \ 1]$: 1st sample is class 2
 2nd sample is class 0
 3rd " class 3
 :

$\Rightarrow \text{np.eye}(4)[Y_{\text{flat}}] = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{matrix} \rightarrow \text{1st sample is class 2} \\ \rightarrow \text{2nd " " 0} \\ : \end{matrix}$ $\therefore$ 각 Row를 sample 1회로 변환

• initialize _ parameters

1. layer_dims?

$\rightarrow$ shape of X, Y

If $\text{layer_dims} = [3092, 50, 20, 10]$ $\longrightarrow$ input layer : 3092 node

hidden layer 1 : 50 node

hidden layer 2 : 20 node

output layer : 10 nodes

2. $L = \text{len}(\text{layer_dims}) - 1$

즉 input, hidden 1, hidden 2, output $\longrightarrow$ 총 4개

However, input layer is excluded. $\therefore L = 3$

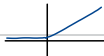
Note He Initialization is used (check Coursera, weight initialization)

• forward _ prop

$$z = W \cdot A + b$$

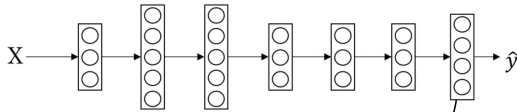
• ReLU, softmax

Sigmoid : $\frac{1}{1 + e^{-x}}$, 

ReLU : $\max(0, z)$, 

Softmax : $\frac{e^{z_i}}{\sum_j e^{z_j}}$, 

Softmax



$z = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is calculate

$$\Rightarrow A = \text{softmax}(z) = \begin{bmatrix} \frac{e^2}{e^2 + e^1 + e^{0.1}} \\ \frac{e^1}{e^2 + e^1 + e^{0.1}} \\ \frac{e^{0.1}}{e^2 + e^1 + e^{0.1}} \end{bmatrix} = \begin{bmatrix} 0.66 \\ 0.24 \\ 0.10 \end{bmatrix} : \text{probability}$$

• Compute Cost

$$\left\{ \begin{array}{l} \text{binary (0 or 1)} : - (y \log(a) + (1-y) \log(1-a)) \\ \text{multi-class} : \mathcal{L}(y, \hat{y}) = - \sum_k y^{(k)} \cdot \log \hat{y}^{(k)} \end{array} \right.$$

• backward propagation

$$dW^{[L]} = \frac{1}{n} \cdot dz^{[L]} \cdot A^{[L+1]T}$$

$$db^{[L]} = \frac{1}{n} \cdot \sum dz^{[L]}$$

$$dA_{\text{prev}} = W^T \cdot dz$$

